Statistics lecture incoming

Please stand by

QM1 Statistics: week 7

Tests for Means and Statistical Errors

Welcome!

Up to the stars

Statistical significance

In a galaxy far away lies the planet Vontrafoodle

- Its beautiful beaches made it into a touristic hotspot …
- … until the peace was disturbed by sadistic seagulls. The birds seemed to have an inborn uge to shit on beach visitors' heads
- Seagulls on the planet Earth had the following shit frequency:

$$\mu = 6$$
$$\sigma = 5$$

- Was the species on Vontrafoodle the same? A research team decided to observe a sample of $n = 53$ birds
- Shit frequency is a **quantitative** variable
- So we should conduct a hypothesis test about the population **mean**
  - Reminder: for <u>categorical</u> variables (with 2 levels), we look at <u>proportions</u>
  - For <u>quantitative</u> variables, we look at <u>means</u>

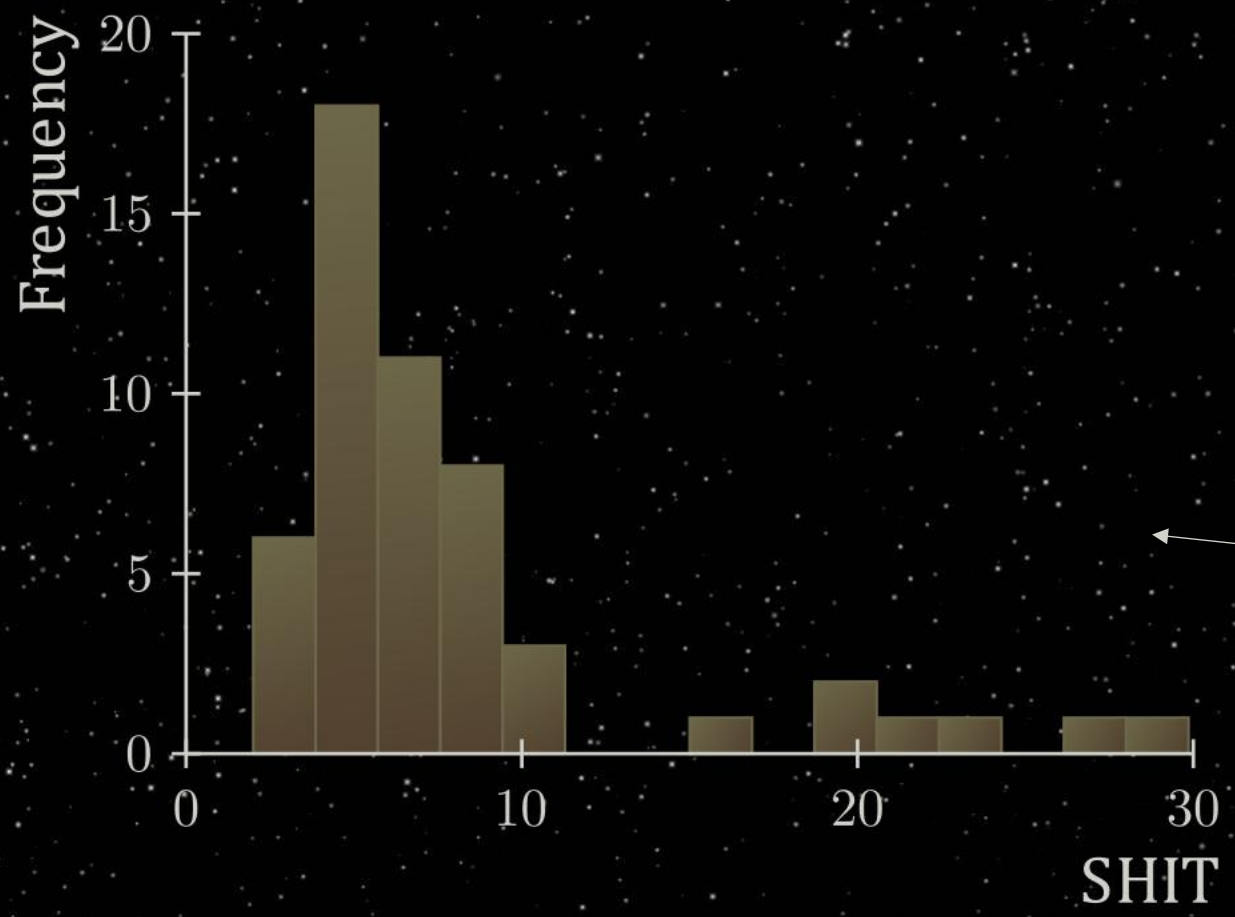
I Hypotheses

- Is there scientific evidence that the average *Larus vontrafoodelis* terrorises beach visitors more often than its Earth ancestors?

$$H_0: \mu = 6$$
$$H_A: \mu > 6$$

- Thus, we shall try to draw a conclusion about the population <u>mean</u> (instead of the population <u>proportion</u>)
- For the significance level α, let's pick the usual 0.05

II Sample results



- $\bar{y} = 8, s = 6.089$
- $\bar{y}$ is more than 6…
  but that could be due to chance!
  We should use this result to test the null hypothesis
- Is the sampling distribution of $\bar{y}$ more or less normal?
  - The distribution of sample scores doesn't look like it came from a normally distributed population (it skews heavily to the right)…
  - But fortunately, *n* is large, so we can rely on the central limit theorem ☺

III Probability calculation

- Remember: we have to estimate the standard deviation using *s*
- For a <u>proportion</u> test, the z-formula was

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$$

- For this <u>mean</u> test, we use the **t-distribution**:

$$t = \frac{\bar{y} - \mu_0}{\frac{s}{\sqrt{n}}}$$

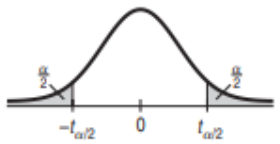
- Fill it all in:

$$t = \frac{8 - 6}{\frac{6.089}{\sqrt{53}}} = 2.391$$

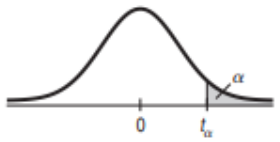
III Probability calculation (continued)

Table T

Values of t_α



Two tails



One tail

Two-tail probability	0.20	0.10	0.05	0.02	0.01	
One-tail probability	0.10	0.05	0.025	0.01	0.005	
df						df
1	3.078	6.314	12.706	31.821	63.657	1
2	1.886	2.920	4.303	6.965	9.925	2
3	1.638	2.353	3.182	4.541	5.841	3
4	1.533	2.132	2.776	3.747	4.604	4
5	1.476	2.015	2.571	3.365	4.032	5
6	1.440	1.943	2.447	3.143	3.707	6
7	1.415	1.895	2.365	2.998	3.499	7
8	1.397	1.860	2.306	2.896	3.355	8
9	1.383	1.833	2.262	2.821	3.250	9
10	1.372	1.812	2.228	2.764	3.169	10
11	1.363	1.796	2.201	2.718	3.106	11
12	1.356	1.782	2.179	2.681	3.055	12
13	1.350	1.771	2.160	2.650	3.012	13
14	1.345	1.761	2.145	2.624	2.977	14
15	1.341	1.753	2.131	2.602	2.947	15
16	1.337	1.746	2.120	2.583	2.921	16
17	1.333	1.740	2.110	2.567	2.898	17
18	1.330	1.734	2.101	2.552	2.878	18
19	1.328	1.729	2.093	2.539	2.861	19
30	1.310	1.697	2.042	2.457	2.750	30
32	1.309	1.694	2.037	2.449	2.738	32
35	1.306	1.690	2.030	2.438	2.725	35
40	1.303	1.684	2.021	2.423	2.704	40
45	1.301	1.679	2.014	2.412	2.690	45
50	1.299	1.676	2.009	2.403	2.678	50
60	1.296	1.671	2.000	2.390	2.660	60
75	1.293	1.665	1.992	2.377	2.643	75
100	1.290	1.660	1.984	2.364	2.626	100
120	1.289	1.658	1.980	2.358	2.617	120

p

t

$df = n - 1 = 52$ (rounded down)

- Now we should look up the p-value
- Crap… where's $t = 2.391$?
  - Just approximate it: it's between 2.009 and 2.403
- So apparently, the p-value is between 1% and 2.5%:

$$0.01 < p < 0.025$$

- Significant… but barely
  - We can reject the null hypothesis

Recall what our p-value represents

Things it is NOT

- $P(H_0)$
  the probability that the null hypothesis is true
- $P(H_0|t = 2.391)$
  the probability that the null hypothesis is true, given the data (the seagulls in our sample)
- $P(t = 2.391)$
  the probability that we would draw a sample of seagulls like this one

What it actually is

- $\boldsymbol{P(t \geq 2.391|H_0)}$
  the probability that this sample of seagulls would be drawn (or an even more unlikely one), if the null hypothesis were true
- So on its own, the p-value doesn't say a thing about the null hypothesis itself (only about the data you got)
- At best, it may tell you that $P(t \geq 2.391|H_0)$ is very small… so maybe $P(t \geq 2.391|H_A)$ is larger?
  (in other words, your sample might have been much more likely under the <u>alternative</u> hypothesis)

The result of a test is **always** uncertain…

- … but sometimes more than otherwise

- $p \leq 0.01$: clearly significant.
  We can reject H_0 with confidence
- $0.01 < p \leq 0.05$: barely significant.
  We can reject H_0, but…
- $0.05 < p \leq 0.10$: just insignificant.
  We cannot reject H_0, but…
- $p > 0.10$: clearly not significant.
  We should not reject H_0 at this time

Be extra aware of a possibly incorrect conclusion: a statistical error

Bugs in the system

Statistical errors

Errors in hypothesis testing

		Test result	
		H_0 not rejected ('Boring birds.')	H_0 rejected ('Holy shit!')
Reality	H_0 true ($\mu = 6$)	Correct decision	Type I error
	H_0 false ($\mu > 6$)	Type II error	Correct decision

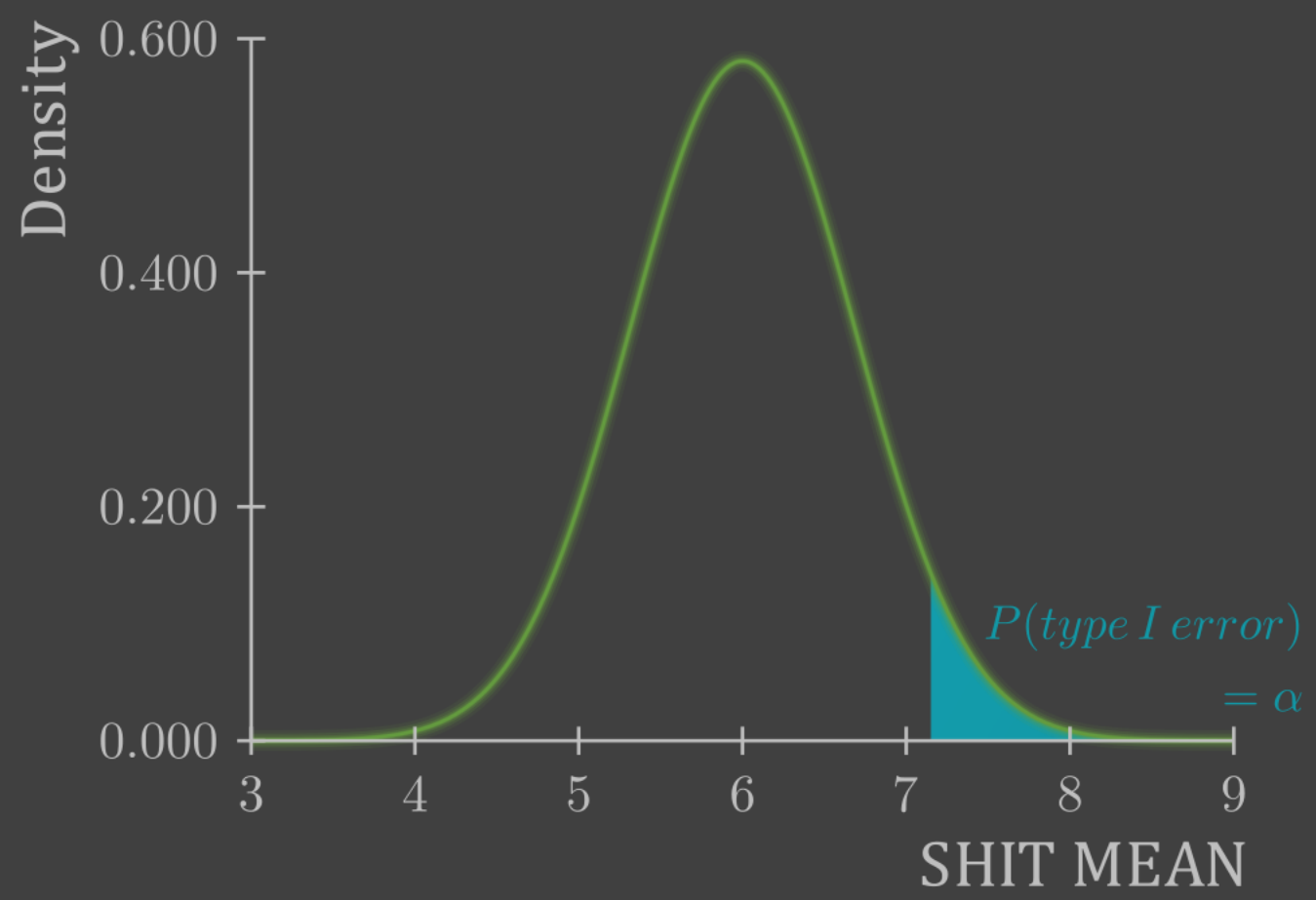
- Statistical tests are not immune to mistakes: sometimes, they will trigger the wrong conclusion about your null hypothesis
- **Type I error:** you falsely reject a true null hypothesis
  - The biologists see a population of common seagulls as special
- **Type II error:** you fail to reject a false null hypothesis
  - The biologists fail to prove how unique the Vontrafoodle gulls are

Errors in hypothesis testing (continued)

		Test result	
		H_0 not rejected ('Boring birds.')	H_0 rejected ('Holy shit!')
Reality	H_0 true ($\mu = 6$)	Correct decision	Type I error
	H_0 false ($\mu > 6$)	Type II error	Correct decision

- The error we can make depends on the state of reality
  - H_0 true? Then the biologists might make a type I error
  - H_0 false? Then they might make a type II error
- But of course, we can never know for sure whether H_0 is true
- So let's account for either scenario

The probability of a type I error



- Always equals the *significance level* (α) that you choose in advance
- Why? Suppose that H_0 was correct and the biologists used $\alpha = 0.05$
- Then by definition, they stood a 5% chance to obtain a significantly high sample mean after all

- So should we just make α as small as possible?
  - Well … this will also make *false* null hypotheses harder to reject
  - More type II errors ☹

The probability of a type II error

- We call this probability β. It depends on multiple things
- β is bigger when:
  - We choose a small α
  - The sample size *n* is small (few birds)
  - There's only a small 'effect size' i.e. small difference between μ_{true} (the real population mean) and μ_0 (what the null hypothesis says about it)
- We often look at it from the other angle: if H_0 is false, how likely is it that we *won't* make a type II error? How likely are we to correctly reject the null hypothesis?
- The probability of a successful rejection is called the **power** of the test $(1 - \beta)$
- The power is bigger when:
  - We choose a large α (not recommended: higher risk of type I errors ☹)
  - The sample size *n* is large (recommended! ☺)
  - There's a large 'effect size' i.e. large difference between μ_{true} (the real population mean) and μ_0 (what the null hypothesis says about it)

Further topics for this week (self-study ☺)

- Sharpe, chapter 13
  - Graphing the power

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture

- **A big thank you to Grant Sanderson (3Blue1Brown) and the Manim community for their amazing Python library!**
  - **Thanks to their groundwork, I was able to make all these animations**

That’s it

After the break: mathematics
Good luck next week! ☺

~ Sir Jordanius - Inter5tellar 5a5uke 5ever